

Installatiehandleiding — RoadAssist

Dit document beschrijft hoe je de RoadAssist applicatie en backend installeert, configureert en opstart op een lokale ontwikkelomgeving.

Platforms: Android (API 26+) en Desktop (Windows en Linux). **Backend:** Ktor server op de JVM.

2. Requirements

Tool	Minimum version
JDK	17
Android SDK	API 26 (build tools for API 36 recommended)
Gradle	8.x (Gradle wrapper included; no separate install required)

No other tools need to be installed manually. The Gradle wrapper downloads all dependencies on first run.

3. Starting the backend

Environment variables

Variable	Required	Description
JWT_SECRET	Yes	HMAC-256 secret used to sign and verify JWT tokens. Set to any strong random string in production. A development fallback is provided in <code>application.yaml</code> ; never use it outside local dev.
DB_MODE	No	<code>sqlite</code> for a persistent file database; omit (or set to <code>h2</code>) for H2 in-memory (data resets on restart).
DB_PATH	No	Path to the SQLite database file. Required when <code>DB_MODE=sqlite</code> . Example: <code>/var/data/roadassist.db</code> .

Run

```
# In-memory H2 database (local dev, data resets on restart)
JWT_SECRET=changeme ./gradlew :server:run

# Persistent SQLite
DB_MODE=sqlite DB_PATH=./roadassist.db JWT_SECRET=changeme ./gradlew :server:run
```

The server starts on port **8080** by default.

4. Building and running on Android

Connect a device or start an emulator; then:

```
./gradlew :app:androidApp:installDebug
```

The app is installed and launched automatically.

Server URL: By default the app points to `https://roadassist.koenv.dev`. To connect to a local server, add a `server_url` string resource in `app/androidApp/src/main/res/values/strings.xml`:

```
<string name="server_url">http://10.0.2.2:8080</string>
```

Use `10.0.2.2` for an Android emulator connecting to localhost, or your machine's LAN IP for a physical device.

5. Running on Desktop

```
./gradlew :app:desktopApp:run
```

This targets the current OS (Windows or Linux). To point the desktop app at a local server, pass the system property:

```
./gradlew :app:desktopApp:run --args='-Droadassist.serverUrl=http://localhost:8080'
```

Or set it in the JVM args inside `desktopApp/build.gradle.kts`. Without this, the app also defaults to `https://roadassist.koenv.dev`.

6. Seed accounts

The server creates two accounts on first startup.

Username	Password	Role
user	user123	Road user
dispatcher	dispatch123	Dispatcher

Passwords are stored as bcrypt hashes. These credentials are for local development only.

7. Running tests

```
./gradlew :core:jvmTest :server:test :app:shared:jvmTest
```

All tests run on the JVM; no Android device or emulator is required.